Epilepsy

Overview

Epilepsy is a chronic neurological disorder characterized by recurrent, unprovoked seizures caused by abnormal electrical activity in the brain. A seizure is a sudden burst of uncontrolled electrical signals that temporarily disrupt normal brain function. Depending on the area of the brain involved, seizures may affect movement, sensation, awareness, behavior, or consciousness.

Epilepsy can develop at any age but is more commonly diagnosed in children and older adults. A person is generally diagnosed with epilepsy after experiencing two or more unprovoked seizures occurring more than 24 hours apart or after one seizure with a high risk of recurrence.

Although epilepsy is a long-term condition, many individuals achieve good seizure control through medication, lifestyle modifications, or surgical treatment. Proper diagnosis and adherence to treatment can significantly improve quality of life.

Types of Epilepsy and Seizures

Epileptic seizures are broadly classified based on where they begin in the brain.

1. Focal Seizures

Focal seizures begin in one specific area of the brain.

They may occur with preserved awareness or impaired awareness.

Common symptoms include:

- Jerking of one arm or leg
- Tingling sensations
- Visual disturbances
- Sudden emotional changes
- Lip smacking
- Repetitive hand movements
- Confusion after the seizure

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2. Generalized Seizures

Generalized seizures involve both sides of the brain from the beginning.

Common types include:

- Tonic-clonic seizures
- Absence seizures
- Myoclonic seizures
- Tonic seizures
- Atonic seizures
- Clonic seizures

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3. Tonic-Clonic Seizures

Previously known as grand mal seizures, these are the most recognizable seizure type.

Symptoms include:

- Sudden loss of consciousness
- Body stiffening
- Rhythmic jerking of the arms and legs
- Tongue biting
- Loss of bladder control
- Confusion and drowsiness after the seizure

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4. Absence Seizures

Absence seizures are more common in children.

Symptoms include:

- Brief staring episodes
- Sudden pause in activity
- Eyelid fluttering

- Immediate recovery afterward

These seizures often last only a few seconds.

Causes

Epilepsy has many possible causes, although in many patients no specific cause can be identified.

Common causes include:

- Genetic factors
- Birth-related brain injury
- Head trauma
- Stroke
- Brain tumors
- Brain infections
- Meningitis
- Encephalitis
- Developmental disorders
- Previous brain surgery

Risk Factors

Several factors increase the likelihood of developing epilepsy.

These include:

- Family history of epilepsy
- Premature birth
- Brain injury
- Stroke
- Dementia

- Brain infections
- High fever causing prolonged seizures during childhood

Symptoms

Symptoms vary depending on the type of seizure.

Common symptoms include:

- Convulsions
- Muscle stiffness
- Sudden jerking movements
- Loss of consciousness
- Blank staring
- Temporary confusion
- Memory loss after the seizure
- Sudden falls
- Unusual smells or tastes before a seizure
- Tingling sensations
- Loss of awareness
- Lip smacking
- Repetitive movements
- Fatigue after the seizure

Some individuals experience an **aura**, which is a warning sign before the seizure begins.

Aura symptoms may include:

- Strange smells
- Visual flashes
- Dizziness
- Anxiety

- Deja vu sensation
- Numbness
- Tingling

Diagnosis

Diagnosing epilepsy involves determining whether seizures have occurred and identifying their underlying cause.

Healthcare providers may perform:

- Detailed medical history
- Neurological examination
- Electroencephalogram (EEG)
- Magnetic Resonance Imaging (MRI)
- Computed Tomography (CT) scan
- Blood tests
- Video EEG monitoring

EEG is one of the most important investigations because it records the electrical activity of the brain and may detect abnormal brain waves associated with epilepsy.

Treatment

Treatment aims to prevent seizures while minimizing medication side effects.

Anti-Seizure Medications

Most patients are treated with anti-epileptic drugs (AEDs).

Examples include:

- Levetiracetam
- Sodium valproate
- Lamotrigine
- Carbamazepine

- Phenytoin
- Oxcarbazepine

The choice of medication depends on seizure type, patient age, pregnancy status, and other medical conditions.

Surgical Treatment

Surgery may be considered for patients whose seizures cannot be controlled with medication and whose seizures arise from a well-defined area of the brain.

Examples include:

- Resective brain surgery
- Laser ablation

Neuromodulation

Some patients benefit from implanted devices such as:

- Vagus Nerve Stimulation (VNS)
- Responsive Neurostimulation (RNS)
- Deep Brain Stimulation (DBS)

These therapies help reduce seizure frequency in selected patients.

Dietary Therapy

Certain children with difficult-to-control epilepsy may benefit from a ketogenic diet under medical supervision.

The diet is high in fat and very low in carbohydrates and may reduce seizure frequency in some patients.

Seizure First Aid

If someone experiences a seizure:

- Stay calm.
- Protect the person from injury.
- Remove nearby dangerous objects.
- Turn the person onto their side after the seizure if possible.
- Loosen tight clothing around the neck.
- Time the seizure.
- Do not place anything inside the person's mouth.
- Do not try to restrain the person's movements.
- Stay with the person until they are fully awake.

Emergency medical care should be sought if:

- The seizure lasts longer than five minutes.
- Multiple seizures occur without recovery.
- The person has difficulty breathing afterward.
- The seizure occurs in water.
- The person is pregnant.
- The seizure results in serious injury.
- It is the person's first seizure.

Lifestyle Management

Patients with epilepsy can often lead active lives with appropriate treatment.

Helpful recommendations include:

- Take medications exactly as prescribed.
- Get adequate sleep.
- Avoid excessive alcohol.
- Manage stress.
- Exercise regularly.

- Wear medical identification if appropriate.
- Attend regular follow-up appointments.
- Avoid known seizure triggers.

Patients should discuss driving, swimming, and operating heavy machinery with their healthcare provider.

Possible Complications

Potential complications include:

- Physical injuries during seizures
- Memory problems
- Depression
- Anxiety
- Medication side effects
- Status epilepticus
- Sudden Unexpected Death in Epilepsy (SUDEP)

Regular medical care helps reduce these risks.

Patient Advice

Patients should never stop anti-seizure medications without consulting their healthcare provider, even if seizures have been controlled for a long time.

Family members should learn basic seizure first aid and know when emergency medical care is required. Maintaining a seizure diary that records seizure frequency, duration, possible triggers, and medication adherence can help healthcare providers optimize treatment.

Summary

Epilepsy is a chronic neurological disorder characterized by recurrent seizures caused by abnormal electrical activity in the brain. Seizures may present in many different ways, ranging from brief staring episodes to full-body convulsions. Diagnosis typically involves neurological examination, EEG, and brain imaging. Most patients achieve good seizure control with anti-seizure medications, while some may benefit from surgery or neuromodulation therapies. Proper treatment, lifestyle modifications, and adherence to medication allow many individuals with epilepsy to lead healthy and productive lives.